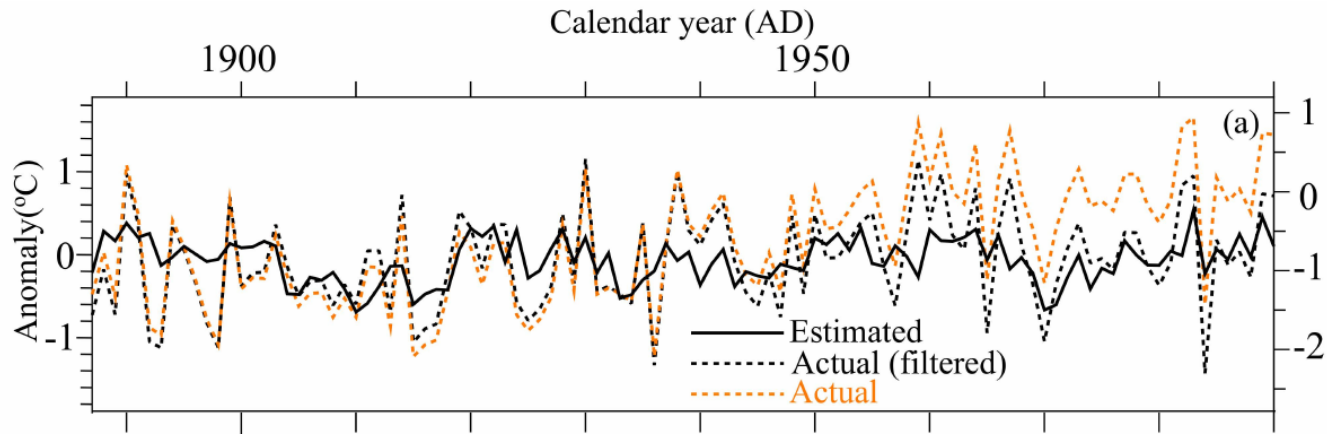
Lesson 7 Read the axes and legend in an original figure

For slide 27. Use this enlargement in print. Ohyama et al. (2013), Figure 3(a), concerns March–May temperature in northeastern Asia.



Panel (a), cropped from the original. Horizontal axis: calendar year. Both vertical axes use °C, but represent differently processed series and different scales.

Ohyama et al., 2013, Fig. 3(a). CC BY 3.0.

Line	Axis and meaning
Solid black	Left: estimated temperature
Dotted black	Left: filtered observations
Dotted orange	Right: unfiltered observations

Identify the appropriate axis before comparing values. Does agreement in this panel establish predictive performance for unseen years?

This is the 1887–1990 calibration fit, not an independent validation plot. Vertical separation of lines on different axes is not directly a temperature difference.

Ohyama et al. (2013), Fig. 3(a), CC BY 3.0

Lesson 7 What does CE compare?

In one sentence

The coefficient of efficiency, CE, measures the reduction in squared prediction error relative to predicting the verification-period observed mean in every year.

$CE = 1 - \Sigma(\text{observed} - \text{predicted})^2 / \Sigma(\text{observed} - \text{verification mean})^2$

CE	Interpretation
1	Perfect predictions for these verification data
0	Same squared error as the verification-mean benchmark
Negative	Larger squared error than that benchmark

The mean is a retrospective benchmark used to assess predictions; it is not assumed to be known when forecasting unseen years. The expression is undefined if the observed verification series is constant.

Table 2 reports split-period CE values of 0.019 and −0.040. Do both support a stable reconstruction? Distinguish the roles of Figure 3(a) and Table 2.

Check

The first represents only a small improvement over the benchmark; the second performs worse. A convincing calibration plot cannot substitute for independent testing. Consider periods, processing and validation design. The processed series in this study also cannot directly recover the long-term warming trend.

Ohyama et al. (2013), Table 2